Classroom

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Assignment - 1

Section - A

Ans 1 Abstraction is the process at hiding complex implementation details and showing only essential features at a System.

We use It because:

→ Reduces system complexity.
→ Makes design and development easier
→ Enables Modularity.

Ans 2 levels of Abstractions

1. Hardware level - Physical component (CPU, memory)
2. Machine level - A Praty ~~Prstw~~ instructions (0s and
3. Assembly level - Symbol ic instructions (ADD, MOV)
4. High-level language - C, Java, Python
5. Application level - End user Programs.

Example

Printing 'Hello' → Python code → Compiled to assembly → Converted to machine code → executed by hardware.

Ans 3 Functional Units

- Input Unit : Takes input from User
- Memory Unit stores data and instructions
- Arithmetic logic Unit (ALU) Performs Calculations
- Control Unit (CU) Control executions of instructions.
- Output Unit

Ans-4 ALU vs CIU

ALU - Performs arithmetics and logical operations

CPU - Controls and Coordinates all operations

Register vs Cache :-

- Registers - Smallest fastest storage inside CPU
- Cache - slightly larger, stores frequently used data

Ans-5 Von - Neumann Model

- A Computer architecture where CPU memory and I/O are Connected : and instructions data share the Same memory.

Stored Program Concept

Instruction are stored in memory like data and executed sequentially

Ans-6 limitation caused due to a single path (bus) between CPU and memory.

Key Points

Data and instructions store same bus
Causes delay in execution
Limits System Performances.

Ans-7 Number of transistions on a chip doubles approximately energy us 24 months

Impact

Increased performances
Reduced Cost
Smallest and more powerful devices.

Ans-8 IPC (instruction Per Cycles):

- IPC is the number of instructions a CPU one cycle.

- Higher IPC - Better CPU Performances.

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Ans-9 Execution Time Total time to run a programs

Clock Rate : Number of Cycle Per Second

CPI (Cycles Per instruction) Avg Cycle Per instruction

IPC Instruction Per Cycle

Relations Execution Time = (Instruction x CPI) / Clock Rate

IPC = 1/CPI

Ans-10 Definations

BPt : Smallest unit (0 or 1)
Byte : 8 bits
Signed : Represents Positive and negative value
Unsigned : Only Positives Values.

Examples

8 bit Unsigned : 0 to 255
8 bPt Signed 128 to +127

-11 Basic operations are:

Arithmatics (addition, Subtraction, Multiplications)
logical (AND, OR, NOT)

Comparison (>, <, =)
Data Movement (load / store)

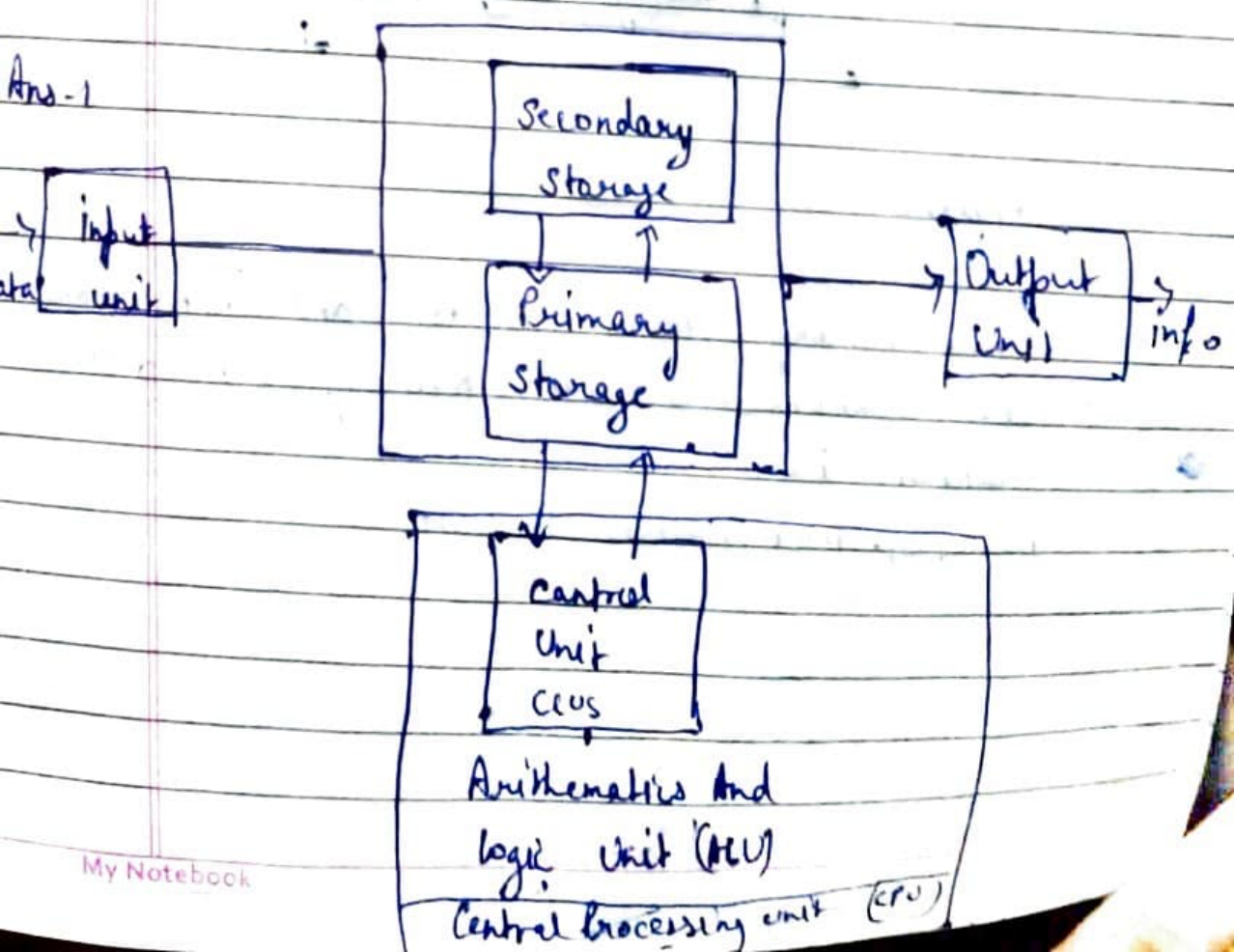
Ans-12 Character Representation

- ASCII (7/8 -bit)
  Unicode (supports Multiple languages)

Decimal Representations
- Fixed -points simple representation)
- Floating -points (Scientific, eg. IEEE 754)

Section B

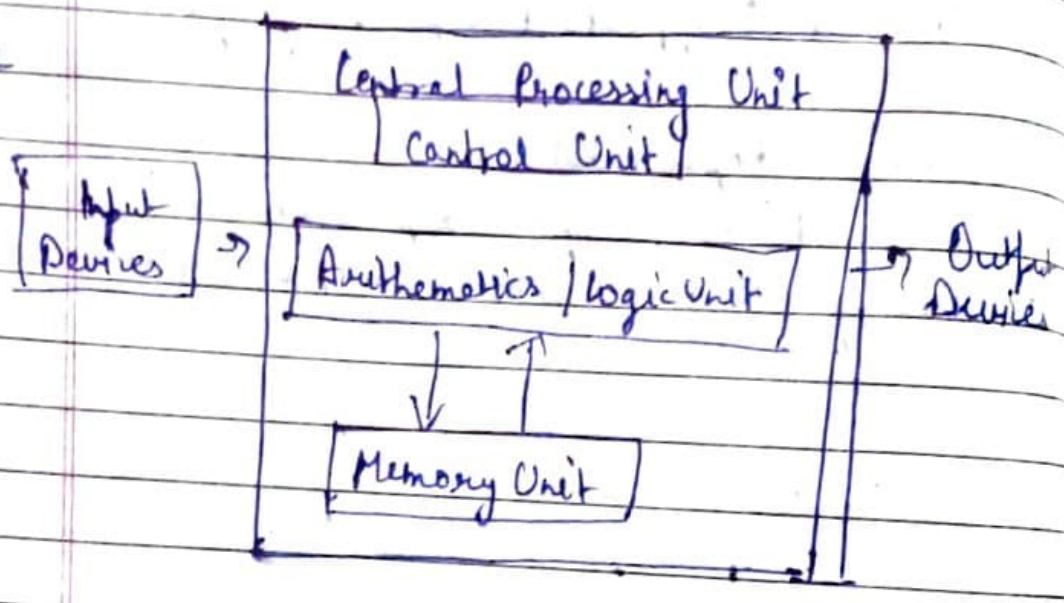
Ans-1



Explanation

- Input unit takes data, CPU Processes it using ALU and CU
- Memory stores data and instructions
- Output unit display results Connects all unit

Ans 2



explanations

→ Single memory stores both data and instructions
→ CPU communicates with memory and I/O via common bus.
→ Basic of most modern computers

PTO

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Ans-3

Instruction Count
↓
CPI
↓
Clock Rate

CPU Execution Time

IPC = 1/CPI

Explanations

- CPU Time depends on Instructions Count, CPI and clock rate
- Higher clock rate ↓ time higher CPI ↑ time.
- IPC is inverse of CPI (higher IPC = better performance

Ans-4 Diagram (Range):

Decimal	-8	-7	-6	-5	-4	-3
	1000	1001	1010	1011	1100	1101

	-2	-1	0
	1110	1111	0000

1	2	3	4	5	6	7
0000	0001	0010	0010	0011	0100	0101

Explanation

4-bit 2's complement range -8 to +7
MSB (leftmost bit) indicates sign
Used to represent negative numbers efficiently

* Section - C

Ans-1 Given

CPU A : 3 GHz, CPI 1.5
CPU B : 2.2 GHz, CPI 1.0

Formula : CPU Time = (Instructions Count × CPI) / clock Rate

Now

CPU A = $\frac{1.5}{3.0}$ = 0.5

CPU B time = $\frac{1.0}{2.1}$ = 0.45

∴ CPU B has lower executions time, hence faster
↳ Reason : lower CPI out weights lower clock rate

Ans-2 Effect of Cache MPs:

- Cache miss → CPU waits (Stall Cycles increase)
- IPC decreases - fewer instructions Per Cycles
- Execution time increases

Fix for it Carry One.
Increases Cache size
Improved cache locality
Use Multi level cache

Ans-3 Given

Range : 200 Values.
Precision : 0.1 - need decimal accuracy.

Step-1 Choice
Use Fixed Point Representation

Step - 2 Justification

- Fixed Point is so sufficient for limited large
- More efficient and faster floating point
- less memory usage and simple hardware

Ans-4 Critiqie of given Statement are mentioned below :

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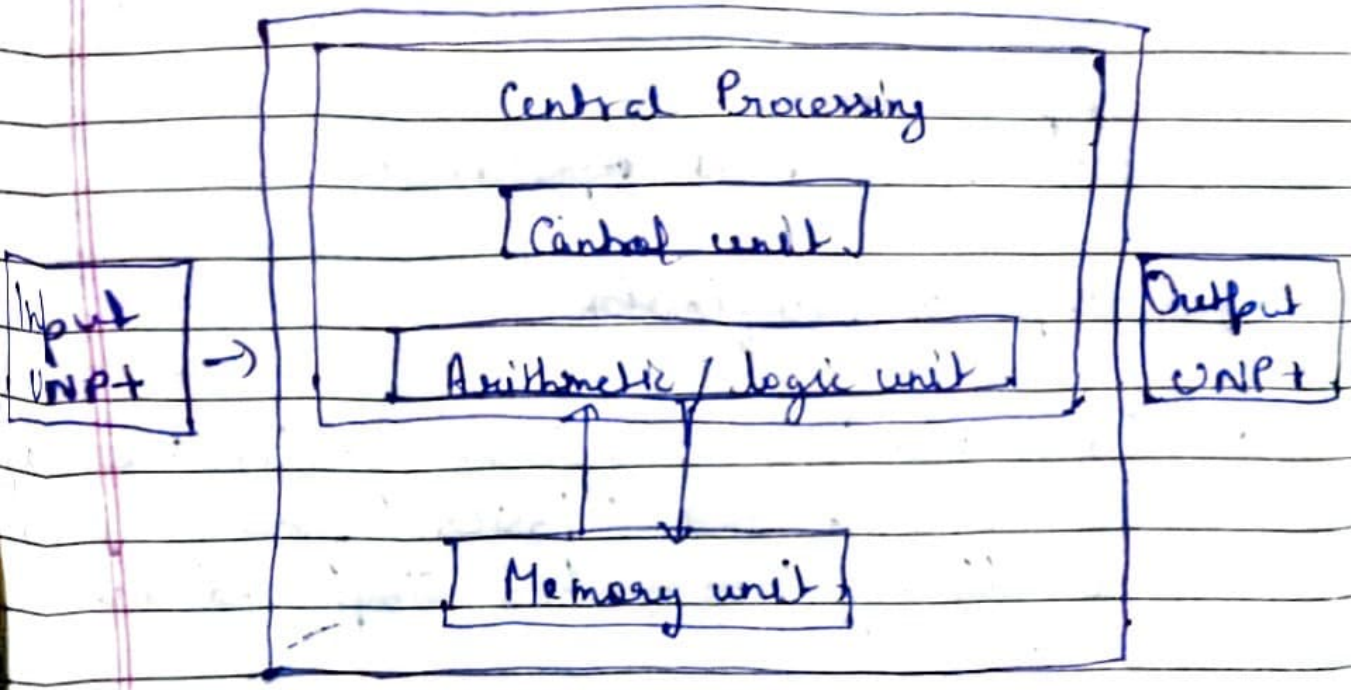
→ Moore's law refers to transistor count not

→ Performance gains now limited due to
- → Power and heat consistents
- → slower clock speed scaling.

→ Modern Perfo improvement depends on
- → Parallelism (multi core - CPUs)
- Better architecture, not

Conclusion Moore's law alone does not gurantee automatics performance doubling today.

Section D



* Von Nevmam Architecture

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Design Decision

Single memory for data and instructions

Explanation

Simplifice System design
Both leads to Non Newmann bottleneck due shared bus.

Ans-2 Performance Relations

CPU Time = (Instruction (Count * CPI) / (Click rate

Explanations
- Performance improves when CPU time decreases
- Depends On IC, CPI and clock Rate

One Practical Improvement

- Reduce CPI using pipeline or better Cache
- Reduce IC using Optimized algorithms
- Increase clock Rate using hardware.

Ans-3 Represent Number

let's take +5 and +4 (4-bit system)

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→ +5 → 0101
→ +4 → 0100

Additions

$$\begin{array}{r} 0101 \\ 0100 \\ \hline 1001 \end{array}$$

Interpretation

1001 in 2's complement = -7

Overflow check

Adding two positive numbers - result should be Positive
But result is negative - Overflow Occurs

In conclusion: Overflow happens when result exceeds representables range.